

Menstrual Cramps

(Dysmenorrhea; Painful Periods)

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Menstrual cramps are pains in the lowest part of the abdomen (pelvis), a few days before, during, or after a menstrual period. The pain tends to be most intense about 24 hours after periods begin and to subside after 2 to 3 days. The pain is often crampy or a dull, constant ache, but it may be sharp or throbbing and may come and go. It sometimes extends to the lower back or legs.

Many women also have a headache, nausea (sometimes with vomiting), and constipation or diarrhea. They may need to urinate frequently.

Some women with painful periods also have symptoms of [premenstrual syndrome](#) (such as irritability, nervousness, depression, fatigue, and abdominal bloating). These symptoms may persist during part or all of the menstrual period.

Dysmenorrhea interferes with daily activities in 30% or more of women and may result in absence from school or work. Women with dysmenorrhea have an increased likelihood of having chronic pelvic pain or other chronic pain conditions.

Causes of Menstrual Cramps

Menstrual cramps may:

- Have no identifiable cause (called primary dysmenorrhea)
- Result from another disorder (called secondary dysmenorrhea)

Primary dysmenorrhea usually starts during adolescence and may become less severe with age and after pregnancy. It is more common than secondary dysmenorrhea.

Secondary dysmenorrhea usually starts during adulthood (unless it is caused by a birth defect of the reproductive system).

Common causes

Experts think that primary dysmenorrhea may be caused by release of substances called prostaglandins into the blood or tissues during menstruation. Prostaglandin levels are high in women with primary dysmenorrhea. Prostaglandins may cause the uterus to contract (as occurs during labor), reducing blood flow to the uterus. These contractions can cause pain and discomfort. Prostaglandins also make nerve endings in the uterus more sensitive to pain.

Anxiety may also contribute to the pain.

Risk factors for severe symptoms include the following:

- Early age of first period
- Long or heavy menstrual periods
- Never having been pregnant
- Age < 30
- Depression, anxiety, or high levels of stress
- Smoking or excess alcohol
- A family history of dysmenorrhea

Symptoms tend to lessen with increasing age and after a first pregnancy.

Secondary dysmenorrhea is caused by abnormalities in the reproductive system. It is commonly caused by

- **Endometriosis**: Tissue that normally occurs only in the lining of the uterus (endometrial tissue) appears outside the uterus. Endometriosis is the most common cause of secondary dysmenorrhea.
- **Adenomyosis**: Endometrial tissue grows into the wall of the uterus, causing it to enlarge and swell during menstrual periods.
- **Fibroids**: These noncancerous tumors are composed of muscle and fibrous tissue and grow in the uterus.
- **Cervical stenosis**: The passageway through the cervix (from the vagina to the main body of the uterus) is narrow or completely closed.

Less common causes

There are many less common [causes of secondary dysmenorrhea](#). They include

- Cysts and tumors in the ovaries
- Use of an [intrauterine device](#) (IUD)
- [Pelvic inflammatory disease](#)
- Scar tissue in the uterus
- Birth defects of the vagina, cervix, or uterus

IUDs that release copper are often associated with painful periods. Those that release a progestin (a synthetic form of the female hormone progesterone) usually do not cause painful periods.

In a few women, pain occurs because the passageway through the cervix (cervical canal) is narrow. A narrow cervical canal (cervical stenosis) may develop after a procedure, as when a precancerous condition (dysplasia) of the cervix is treated.

Evaluation of Menstrual Cramps

Doctors usually diagnose dysmenorrhea when a woman reports that she regularly has bothersome pain during menstrual periods. They then determine whether dysmenorrhea is primary or secondary.

Warning signs

In women with dysmenorrhea, certain symptoms are cause for concern:

- Severe pain that began suddenly or is new
- Constant pain
- Fever
- A pus-like discharge from the vagina
- Sharp pain that worsens when the abdomen is touched gently or the person moves even slightly

When to see a doctor

Women with any warning sign should see a doctor as soon as possible (usually the same day).

If women without warning signs have more severe cramps than usual or have pain that lasts longer than usual, they should see a doctor within a few days.

Other women who have menstrual cramps should call their doctor. The doctor can decide how quickly they need to be seen based on their other symptoms, age, and medical history.

What the doctor does

Doctors or other health care professionals ask about the pain and the medical history, including the menstrual history. Clinicians then do a physical examination. What they find during the history and physical examination may suggest a cause of menstrual cramps and the tests that may need to be done (see table

Some Causes and Features of Menstrual Cramps

For a complete menstrual history, clinicians ask women to describe their periods:

- Age when menstrual periods started
- How many days they last
- How heavy they are
- How long the interval between periods is
- Whether periods are regular
- Whether spotting occurs between periods
- When symptoms occur in relation to periods

Clinicians also ask women to describe the following:

- Age when menstrual cramps began
- Any other symptoms, including pelvic pain unrelated to periods
- Severity of the pain, what relieves or worsens symptoms (including nonsteroidal anti-inflammatory drugs), and how symptoms interfere with her daily activities
- Any pain during sexual activity or problems becoming pregnant (as may occur in endometriosis)

Doctors ask about other disorders and conditions that can cause cramps, including use of certain drugs (such as birth control pills) or an IUD. Doctors ask about any surgical procedures that increase the risk of pelvic pain, such as a procedure that destroys or removes the lining of the uterus (endometrial ablation).

A pelvic examination is done. Doctors check for abnormalities, including discharge, pelvic area tenderness or enlargement of the uterus, cervical polyps, and fibroids.

Doctors also gently feel the abdomen to check for areas of particular tenderness, which may indicate severe inflammation in the abdomen (peritonitis).



Some Causes and Features of Menstrual Cramps

Cause	Common Features*	Tests
Adenomyosis (growth of tissue that normally lines the uterus—called endometrial tissue—into the wall of the uterus)	Heavy, painful periods, vaginal bleeding between periods, pain in the lowest part of the abdomen (pelvis), and a feeling of pressure on the bladder and rectum Sometimes pain during sexual intercourse	Ultrasound or MRI of the pelvis In women with abnormal vaginal bleeding, sometimes a biopsy to rule out other causes of bleeding
Birth defects of the reproductive tract (rare)	Sometimes periods that never start (primary amenorrhea) Sometimes abnormalities found during the pelvic examination, such as a blockage in the vagina or a lump in the pelvis	Sometimes hysterosalpingography (x-rays taken after a contrast agent is injected into the uterus and fallopian tubes), sonohysterography (ultrasound after fluid is infused into the uterus), or MRI
Cervical stenosis (narrowing of the passageway through the cervix)	Irregular or no menstrual periods, vaginal bleeding between periods, infertility, and abdominal pain that occurs in cycles	Sometimes ultrasound of the pelvis
Cysts and tumors in the ovaries (ovarian cancer or noncancerous ovarian growths)	Often no other symptoms Sometimes abnormal vaginal bleeding	Ultrasound or other imaging tests If cancer is suspected, blood tests to measure substances



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